

2. Inos is a Welsh caravan manufacturer based in Denbighshire. The materials used to build modern caravans are very different to those used in early caravans.

Table 1 shows how the materials used in manufacturing the body of a caravan have changed over time.

Year	Materials used
1920	wood
1950	steel and wood
1970	aluminium and wood
2015	glass reinforced plastic (GRP), polyester

Table 2 shows information about some of the materials used to make the body of caravans.

Material	Density (g/cm ³)	Stiffness (GPa)	Melting Point (°C)	Strength (MPa)	Is it brittle?	Does it corrode?
wood	0.6			1 000	No	no but rots when wet
steel	7.8	210	1 357	1 200	No	yes producing rust
aluminium	2.7	69	660	90	No	yes producing a resistant coating
GRP	glass fibres	2.4	1 400	3 500	Yes	No
	epoxy resin	1.5		60	Yes	No
polyester	1.9	150	121	250	Yes	No

Use the information above to answer the following questions.

- (a) Describe how the weight and strength of caravan bodies changed between 1920 and 1970. [4]

Weight:

.....

.....

Strength:

.....

.....

